CTRL: A Joint Framework for Health-Aware Cluster-Head Selection Cooperative Task Offloading and Worker UAV Selection in FANETs

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***Abstract*—Flying ad hoc networks (FANETs), struggle to keep tasks running reliably because the UAVs move frequently, have limited battery power,deal with changing wireless links, and get tasks arriving at different rate in various groups.Existing methods for UAV clustering and task offloading handle Cluster Head(CH) task offloading, and UAV redeployment as separate problems. This separation cause some cluster to become overloaded, reduces resource utilization, and increases service delay. This paper presents CTRL, a combined system that handles choosing cluster leaders, moving tasks to other locations, and repositioning UAVs in FANETs.CTRL chooses a cluster head by using a model that looks at how much energy is left, how hot the device is, how well the connections are, and how good the device is at doing calculations. When a cluster gets too loaded, the chosen leader in that cluster starts working with others by picking a helpful UAVs from nearby clusters, but only if it meets certain time and availability rules. CTRL also sends the chosen worker UAV back to the overloaded cluster to help balance the workload over time and keep the tasks run smoothly. The new framework gives a solid theory for choosing cluster heads and brings together offloading and redeployment into one system that helps balance the load. The simulation results indicate that CTRL decreases task delay, enhances energy efficiency, and provides a more even spread of work compared to current methods used for clustering drones and assigning tasks.**

***Index Terms*—Flying ad hoc networks, UAV clustering, Task offloading, Cluster Head selection, Load balancing**

I. Introduction

The rapid growth of unmanned aerial vehicles (UAVs) uses in multiple parts of our surrounding like agriculture, military operations, land surveying, disaster management, and other emerging domains. However, manual control is often insufficient to exploit the full functionality of UAVs, so fully automatic control is required for satisfactory end-user performance. Since users frequently generate and retrieve data, network overloads results, Mobile Edge Computing (MEC) deployed at the UAV alleviates this by offloading computation to edge nodes, lowering latency and improving reliability and connectivity [1]. Both base stations and UAVs possess computing power that can be assigned to offload UAV tasks, but determining the appropriate number of clustering centers to balance energy consumption against clustering overhead remains difficult [2]. Coordinating a UAV swarm without a central authority is tedious, motivating a cluster head in a hierarchical UAV swarm network to act as a communication and coordination center for other UAVs [3]. Similarly, in wireless sensor networks (WSNs), continuous communication among nearby sensor nodes (SNs) causes rapid energy depletion, so a cluster head is needed to coordinate SNs for load balancing and to minimize energy depletion [4].

UAVs are widely used in real-time applications such as disaster management, surveillance, military operations, and package delivery due to their ease of deployment, flexibility, and mobility. UAV swarm networks have gained popularity for collaboratively performing complex tasks more efficiently than a single UAV, despite the limited computation power, storage, and battery imposed by their compact size. Coordinating UAVs within a swarm is tedious, so clustering techniques divide UAVs into multiple clusters, each with a Cluster Head (CH) responsible for task coordination, communication with neighboring cluster heads, and resource allocation. This cluster-based framework reduces communication overhead, improves scalability, and enables efficient inter-cluster communication and stable workload distribution. While task demand stays largely static in static UAV environments, it varies continuously across clusters in dynamic environments, causing some clusters to become overloaded while others remain underutilized. This imbalance increases processing delay, communication overhead, and energy degradation, degrading overall network performance.

UAV redeployment among clusters addresses this imbalance by transferring an underused UAV to an overloaded cluster to facilitate task execution and balance load. However, selecting suitable UAVs for such transfers is challenging given dynamic network conditions, limited energy resources, and the mobility of both UAVs and ground users, raising the question of which UAV should migrate, under what conditions, and how task distribution should be divided efficiently. To address UAV redeployment and load balancing, a non-cooperative game-theoretic approach is adopted.

A. Motivation

Unmanned Aerial Vehicles(UAVs) are widely used in many real world applications because of small sizes of UAVs ,such as disaster management,surveillance and package delivery from warehouses to home.In spite of small size of UAVs which forms swarm UAV network, is a complex task because of network toplology changes,limited resource and frequent movement of UAVs.To tackle these challenges multiple cluster forms,where each cluster have dedicated Cluster Head(CH) which manages UAVs in that particular cluster for resource sharing,user reassignment and energy efficiently. With such cluster system formation by UAV ,Cluster Head also communicate with other cluster to provide facility to redeployment of UAVs from one to another at the time of addition UAV needed when that particular cluster cannot handle with existing UAV causes can be limited resources or load has increased drastically in dyanamic environment. Therefore, combining a clustering mechanism with a non- cooperative game-theoretic approach gives an effective and reliable solution for redeployment of UAV in swarm of network.By integrating those two things scalabilty,reliability improves in UAV swarm networks.

B. Contribution

To deal with changing workloads, unreliable UAVs connections, and limited resources when tasks are executed by UAVs in FANETs, we introduce **CTRL**, a single framework that is aware of system health.This framework works together to choose cluster head,share task cooperatively, and pick worker UAVs for cluster that are overloaded. The main points that this work listed below

- First, We create a method for choosing cluster heads that takes into account the specific working conditions of UAVs, such as remaining energy, temperature, link quality,and computing power. CTRL formulates CH election as non-cooperative game where each UAV chooses its broadcast power to balance SINR based coverage gain with energy expenditure, rather than the existing clustering and CH selection methods that rely mainly on residual energy, connectivity, or learning-based head detection [3], [4].This allows unhealthy, overheated, weakly connected,or low energy UAVs to automatically leave the list of possible CHs without needing a fixed exclusion limit set by someone else.
- Second, Unlike other UAV MEC studies that handle task offloading, trajectory control, and resource allocations as separate issues [1], [2], [16], we combine CH selection and cooperative inter cluster offloading into one process. When a cluster gets overloaded, the chosen CH sends a Help Request Beacon(HRB) that includes details about the task, when it needs to be done, how much energy is required, and where the cluster is located. The nearby cluster that helps is chosen by looking at several things together, like how good the signal is between clusters, how far they are from each other, how many steps it takes to communicate, how many worker drones are available, how much extra power there is, how reliable past tasks were completed, whether the task done on time and how much energy left.
- Third, Within the neighboring cluster chosen, we develop a strategy for selecting worker UAVs that is based on Nash bargaining. Instead of just giving tasks based on leftover energy, distance, or how much computing power is left [1], [2], CTRL looks at each possible worker UAV using a surplus utility that considers their health, remaining CPU speed, how reliable the connection, quality of service, mobility exposure, total time to finish the task and energy feasibility. This ensures that the selected worker UAV improves the overload cluster's service capability while remaining individually rational for the assisting UAV.
- Finally,

II. Related Work

A. UAV Clustering and Cluster-Head Selection

Clustering is commonly used in flying ad hoc networks(FANETs) to improve scalability, reduce communication overhead, and support coordination among mobile UAVs. Mozaffari et al. discuss key challenges in UAV enabled wireless networks. These challenges include dynamic topology, energy efficiency, three dimensional deployment, and reliable communication [7]. Zhang et al.proposed a location and velocity prediction assisted FANET clustering algorithm. This algorithm aims to improve cluster stability in highly mobile UAV networks [6]. Mou et al. developed a graph self supervised learning method for detecting cluster heads(CH) in hierarchical UAV swarms [3]. Energy aware CH selection studied in sensor network environments. This research seeks to reduce energy depletion and extend network lifetime [4]. However, these studies mainly look at mobility connectivity, residual energy, or learning based CH detection. They do not consider residual energy, thermal state, link quality and computation capability together when electing CHs. In contrast, CTRL treats CH selection as a health aware non cooperative game. In this model, each UAV adjusts its broadcast power by balancing SINR based coverage gain and energy use.

B. Task Offloading and Resource Allocation in UAV-Assisted MEC

UAV-assisted mobile edge computing (MEC) extensively studied to minimize task execution latency and enhance computational resources in dynamic wireless networks. Zhao *et al.* introduced a multi agent deep reinforcement learning framewrok for task offloading in UAV supported MEC, where various agents learn offloding choices amidst changing network conditions [1]. Guo *et al.* investigated cooperative task offloading and resource distribution among multiple UAVs in 5G Advanced and beyond, aim to enhance computational efficiency and minimize service delay [2]. Li*et al.* collaboratively enhanced resource distribution and UAV path for energy efficient UAV supported MEC [16]. Wang *et al.* implemented deep reinforcement learning for dyanmic trajectory management in UAV assited MEC amids varying network condition [13]. While these studies enhance task offloading, resource allocation, and trajectory management, they typically optimize these choices once the serving UAV or network confgiuration has already been established. CTRL distinguishes itself from these studies by linking CH electionto cooperative inter cluster offloading and by resewing outputs from the election stage, such as the UAV health index, equilibrium broadcast power and utility score.

C. Load-Aware Inter-Cluster Offloading and Worker UAV Selection

D. Research Gap

Existing studies usually treat CH selection, task offloading, resource allocation, and worker UAV selection as separate modules. This separation can lead to overloaded clusters, inefficient resource utilization, high service delay, and premature energy depletion of important UAVs. CTRL fills this gap by jointly performing health-aware CH election, cooperative inter-cluster offloading, and Nash-bargaining-based worker UAV selection for load balancing in FANETs.

III. System Model

A. Problem Scenario

B. System Architecture

C. Network Topology

Creating a Cartesian coordinate system of a three-dimension system. $\mathcal{N} = \{1, 2, \ldots, N\}$ of UAVs hovers at a unchangeable altitude H (m) above the ground, providing its service to set $\mathcal{G} = \{1, 2, \ldots, G\}$ of User Equipments(UEs) operating by users on ground. In individual time interval $t \in \mathcal{T} = \{1, \ldots, T\}$, the 3-D positioning of the UAV n is

$$\begin{aligned} d_{im} &= \|\mathbf{a}_i - \mathbf{a}_m\| \\ &= \sqrt{(x_i - x_m)^2 + (y_i - y_m)^2 + (h_i - h_m)^2}. \end{aligned} \tag{1}$$

$$\mathbf{q}_n(t) = \big(x_n(t),\, y_n(t),\, H\big), \quad n \in \mathcal{N}. \tag{2}$$

and UE position g is $\mathbf{s}_g(t) = (x_g(t), y_g(t), 0)$. The UAVs makes a swarm so partitioned into K disjoint clusters $\{\mathcal{C}_1, \mathcal{C}_2, \ldots, \mathcal{C}_K\}$ such that $\bigcup_{k=1}^{K} \mathcal{C}_k = \mathcal{N}$ and $\mathcal{C}_k \cap \mathcal{C}_{k'} = \emptyset$ for $k \neq k'$. Except cluster $\mathcal{C}_k$, exactly one among from all the UAVs works as the *cluster head* (CH) and the all others are *cluster members* (CMs).

D. Air-to-Air Communication Model

1) UAV-to-UAV (Intra-Cluster) A2A Channel: in spite of all UAVs control at identical altitude H, the communication link inside intra-cluster are primarly line-of-sight(LoS). cluster member m from UAV i transmitting with power p_i with recieved signal-to-interference plus-noise ratio (SINR) is

$$\mathrm{SINR}_{im} = \frac{p_i g_{im} d_{im}^{-\eta}}{\sigma^2 + \sum_{j \neq i} p_j g_{jm} d_{jm}^{-\eta}} \tag{3}$$

where d_{im} express the inter-UAV distance, g_{im} express the channel gain, η is the path-loss exponent, σ^2 is the noise power, and p_i is UAV transmit power i.

For Line of Sight(LoS) UAV-to-UAV communication, the path-loss exponent usually ranges from 2 to 2.5 is taken from [8]. In this work, $\eta =$ is borrowed.

The gain associated with small-scale fading gain adhere to Rayleigh fading framework and is represented as

$$g_{im} \sim \mathrm{Exp}(1). \tag{4}$$

The thermal noise power is

$$\sigma^2 = F k_B T_{\mathrm{env}} B. \tag{5}$$

where F is noise from reciver side, k_B is standard Boltzmann's constant, T_{env} is the environment temperature, and B is bandwidth of the entire channel is adopted from [8].

The transmit power satisfies

$$0 \leq p_i \leq P_i^{\max}. \tag{6}$$

where $P_i^{\max}$ signifies the maximum permissible transmission power of the UAV i i.

a) Small-scale fading gain g_{im}: The received challenging baseband is

$$\tilde{y}_{im} = \sum_{\ell=1}^{L} a_\ell e^{j\phi_\ell} s(t - \tau_\ell). \tag{7}$$

where a_ℓ, $\phi_\ell \sim \mathcal{U}[0, 2\pi)$, and τ_ℓ are the ℓ-th multipath amplitude, phase, and delay. For L sufficiently large, the Central Limit Theorem gives $\mathrm{Re}[\tilde{y}_{im}], \mathrm{Im}[\tilde{y}_{im}] \sim \mathcal{N}(0, \sigma_h^2/2)$, so the power gain

$$g_{im} = |\tilde{y}_{im}|^2 \overset{d}{=} \mathrm{Exp}(1), \quad \mathbb{E}[g_{im}] = 1 \tag{8}$$

Rayleigh power follow a average channel condition which is 1 in Exp. it is a standard worst case for LoS UAV when Ricin K-factor is not known [8].

b) Transmit power p_i: The power $p_i \in [0, P_i^{\max}]$ is the strategy varying of UAV i for electing CH on applying election game . Its maximum power limit for transmitting P_i^{hw} and the rest energy during a time interval:

$$P_i^{\max} = \min\left(P_i^{\mathrm{hw}}, \frac{E_i^{\mathrm{res}}}{\Delta t}\right) \tag{9}$$

where E_i^{res} (J) is examined by the preinstalled coulomb counter and Δt (s) is the time-interval .

2) Time-slot duration Δt: The channel assumption required $g_{im}$is given by quasi-static to stay steady within each particlar slots. The channel coherence time is given by the Jakes model is

$$T_c = \frac{9c}{16\pi v_i^{\max} f_c} \tag{10}$$

where c is speed of light, $v_i^{\max}$ is the maximum UAV speed, and f_c is the carrier frequency. The outcome follows from the level-crossing-rate analysis of the Rayleigh,which give in Doppler Bandwidth $f_D^{\max} = v_i^{\max} f_c / c$, and hence $T_c \approx 9/(16\pi f_D^{\max})$ [9] The time duration must satisfy

$$\Delta t \leq T_c \quad \text{and} \quad \Delta t \geq T_{\mathrm{meas}} + T_{\mathrm{proc}} \tag{11}$$

where T_{meas} is the pilot-symbol overhead time spent during sending it and T_{proc} is the KKT optimization after getting channel information

Fig. 1: Proposed CTRL system architecture.

E. SINR-Based Coverage Probability

The power of all the UAVs are $\mathbf{p} = (p_1, \ldots, p_{N_k})$,the probabilty that UAV i' brodcast can decoded by CM (i.e., $\mathrm{SINR}_{im} \geq \gamma_{\min}$) is obtained by averaging (3) over the Rayleigh fading distribution (8). Since $g_{im} \sim \mathrm{Exp}(1)$, we have

$$\Pr[\mathrm{SINR}_{im} \geq \gamma_{\min}] = \Pr\left[g_{im} \geq \frac{\gamma_{\min}(\sigma^2 + I_{im})}{p_i d_{im}^{-\eta}}\right]$$
$$= \mathbb{E}_{I_{im}}\left[\exp\left(-\frac{\gamma_{\min}\,\sigma^2\, d_{im}^{\eta}}{p_i}\right) \cdot \prod_{j\neq i} \frac{p_i\, d_{im}^{-\eta}}{p_i\, d_{im}^{-\eta} + \gamma_{\min}\, p_j\, d_{jm}^{-\eta}}\right] \tag{12}$$

where the expectation uses the Laplace transform of $g_{jm} \sim \mathrm{Exp}(1)$ for each interfering UAV j. The expected coverage fraction of all CMs by UAV i is

$$\mathbb{E}[C_i(p_i, \mathbf{p}_{-i})] = \frac{1}{N_k - 1} \sum_{\substack{m\in\mathcal{C}_k \\ m\neq i}} \Pr[\mathrm{SINR}_{im} \geq \gamma_{\min}] \tag{13}$$

1) Derivation of $\gamma_{\min}$: The $\gamma_{\min}$ is not assumed as a constant; it is obtained from the minimum rate required for successful beacon/control-message decoding during the initial CH election stage. When UAVs are scattered, each UAV broadcasts a beacon message to estimate how many neighbouring UAVs can be reliably reached. The required spectral efficiency is determined by the beacon descriptor as

$$R_{\mathrm{req}} = \frac{L_b}{T_b \cdot B_c} \quad \text{(bits/s/Hz)}. \tag{14}$$

where L_b is the beacon/control packet size in bits, T_b is the maximum allowable beacon decoding time, and B_c is the control-channel bandwidth.

Using the Shannon rate relation, the achievable spectral efficiency of the UAV-to-UAV control link is expressed as

$$R_{im} = \log_2\left(1 + \mathrm{SINR}_{im}\right). \tag{15}$$

For reliable decoding, the achievable spectral efficiency must satisfy $R_{im} \geq R_{\mathrm{req}}$. Therefore, the minimum SINR threshold required for successful beacon decoding is obtained as

$$\gamma_{\min} = 2^{R_{\mathrm{req}}} - 1 = 2^{\frac{L_b}{T_b \cdot B_c}} - 1. \tag{16}$$

Thus, $\gamma_{\min}$ is a system-derived parameter that reflects the minimum communication quality required for reliable control-message exchange. A UAV-to-UAV link from candidate CH i to member UAV m is considered reachable only when

$$\mathrm{SINR}_{im} \geq \gamma_{\min} \tag{17}$$

F. Health Index H_i

The overall health of a UAV is shown by $H_i \in [0, 1]$ which examines three independent and non-overlapping aspects of the UAVs operating condition: energy , thermal state, and link quality for communication

1) Energy health h_i^E: Since battery tolerance is a major factor for UAV operation,the health component is derived from the battery behaviour. Most of the research UAVs used Lithium-Polymer(LiPo) batteries,whose voltage performs a nonlinear relationship with charge state.Shepherd electrochemical battery model accurately obtained [10]:

$$V_{\mathrm{batt}}(q) = E_0 - K\frac{Q}{Q - q} + Ae^{-Bq} \tag{18}$$

where E_0 (V) is the open-circuit voltage, K (V/Ah) the polarisation constant, Q (Ah) the rated capacity, A (V) the exponential amplitude, B (1/Ah) the exponential time constant, and q (Ah) the extracted charge—all from the IEC 61960 discharge test. The true *usable* energy remaining from charge state q down to the hardware cutoff voltage V_{cut} is

$$\begin{aligned} E_i^{\text{usable}}(q) &= \int_q^Q \big[V_{\text{batt}}(q') - V_{\text{cut}}\big]\, dq' \\ &= (E_0 - V_{\text{cut}})(Q-q) - KQ\ln\!\left(\frac{Q}{Q-q}\right) \\ &\quad + \frac{A}{B}\left(1 - e^{-B(Q-q)}\right). \end{aligned}$$

The normalized usable energy gives energy health:

$$h_i^{E,\text{exact}}(q) = \frac{E_i^{\text{usable}}(q)}{E_i^{\text{usable}}(0)}. \tag{19}$$

For ease of analysis in the KKT derivation, approximate $h_i^{E,\text{exact}}$ by a smooth exponential $1 - e^{-\alpha_E s}$ with $s = E_i^{\text{res}}/E_i^{\max} \in [0,1]$. The parameter α_E is determined by minimising the L^2 approximation error:

$$\min_{\alpha_E \geq 0} \int_0^1 \Big[h_i^{E,\text{exact}}(s) - \big(1 - e^{-\alpha_E s}\big)\Big]^2 ds \tag{20}$$

Setting the derivative with respect to α_E to zero yields the implicit optimality condition

$$\int_0^1 h_i^{E,\text{exact}}(s)\cdot s\, e^{-\alpha_E s}\, ds = \int_0^1 \big(1 - e^{-\alpha_E s}\big)\cdot s\, e^{-\alpha_E s}\, ds, \tag{21}$$

$$h_i^E = 1 - \exp\!\left(-\alpha_E \frac{E_i^{\text{res}}}{E_i^{\max}}\right) \tag{22}$$

2) Thermal health h_i^T: The onboard thermal sensor provides the current CPU temperature T_i^{curr} (K).since increase in temperature can influence performance throttling ,thermal health is obtained by normalizing T_i^{curr} and $T_i^{\min}$ and the maximum acceptable temperature $T_i^{\max}$ given by

$$h_i^T = 1 - \frac{T_i^{\text{curr}} - T_i^{\min}}{T_i^{\max} - T_i^{\min}} \in [0,1] \tag{23}$$

The CPU frequency of UAV i after inspecting thermal throttling is obtained by:

$$f_i^{\text{res}} = f_i^{\max} h_i^T \tag{24}$$

This establishes a one-way dependency from the current temperature to the thermal health and then to the residual CPU frequency, expressed as

$$T_i^{\text{curr}} \to h_i^T \to f_i^{\text{res}}$$

Therefore, the thermal-health component is independent of the energy-health component (h_i^E), since (h_i^E) is calculated only from the residual battery energy E_i^{res}.

3) Link-quality health h_i^L: The Received Signal Strength Indicator (RSSI) is measure of how well a signal can be received over time,which is different from the immediate signal strength in each slot,called per slot channel chain g_{im}. time-averaged reachability metric, Let RSSI_{im} be the smoothed RSSI value measure from UAV(m) to UAV(i),and let ($\text{RSSI}_{\min} = -90\,\text{dBm}$) be the lowest RSSI level that the hardware can still decode.

$$h_i^L = \frac{1}{|\mathcal{C}_k| - 1} \sum_{\substack{m\in\mathcal{C}_k \\ m\neq i}} \mathbf{1}\big[\text{RSSI}_{im} \geq \text{RSSI}_{\min}\big] \in [0,1] \tag{25}$$

4) Composite index: The three dimensions are combined via a weighted geometric mean:

$$H_i = \big(h_i^E\big)^{w_E}\big(h_i^T\big)^{w_T}\big(h_i^L\big)^{w_L} \tag{26}$$

with $w_E + w_T + w_L = 1$. The geometric (rather than arithmetic) mean ensures that $H_i \to 0$ whenever *any* dimension collapses, reflecting the fact that a UAV with dead batteries, a throttled CPU, or an isolated radio link is ineligible to serve as CH regardless of its other metrics. Weights $(w_E, w_T, w_L) = (0.5, 0.25, 0.25)$ are derived via Saaty's Analytic Hierarchy Process with consistency ratio $\text{CR} < 0.1$.

Non-overlap summary: h_i^E uses E_i^{res}; h_i^T uses T_i^{curr} (different physical sensor); h_i^L uses RSSI (time-averaged link metric, independent of the per-slot g_{im} in (8)).

IV. Cluster Head Election Game

A. Problem Formulation

Within cluster $\mathcal{C}_k$ (cardinality $N_k = |\mathcal{C}_k|$), every UAV competes for the CH role. The CH needs to send coordination messages to every other UAVs.CMs; hence the natural strategy variable in the coordination broadcast power $p_i \in [0, P_i^{\max}]$.every UAV competes for the CH role. The CH needs to send coordination messages to every other UAVs.CMs; hence the natural strategy variable in the coordination broadcast power $p_i \in [0, P_i^{\max}]$.A high value of p_i leads to broader coverage but uses more energy. A UAV that is running low on power or has slowed down because of heat will naturally choose,low power,analytically self-excluding without any externally imposed threshold.

Definition 1 (CH Election Game)**.** *The CH election game is the triple $\mathcal{G}_{\text{CH}} = \langle \mathcal{C}_k, \{[0, P_i^{\max}]\}_{i\in\mathcal{C}_k}, \{\pi_i\}_{i\in\mathcal{C}_k}\rangle$, where $\pi_i : \prod_{j\in\mathcal{C}_k}[0, P_j^{\max}] \to \mathbb{R}$ is the payoff of player i.*

B. Payoff Function

The payoff of UAV i balances coverage gain against energy expenditure, weighted by its health index:

$$\pi_i(p_i, \mathbf{p}_{-i}) = H_i\, \mathbb{E}[C_i(p_i, \mathbf{p}_{-i})] - \frac{(1 - H_i)\, p_i\, \Delta t}{E_i^{\text{res}}} \tag{27}$$

a) Interpretation: When $H_i \to 1$: the UAV is healthy, so coverage gain dominates and the energy cost term vanishes; the UAV competes aggressively. When $H_i \to 0$: the energy-cost coefficient $\to (E_i^{\rm res})^{-1} \cdot \Delta t$, making π_i strictly decreasing in p_i, so the optimal power is $p_i^* = 0$—the UAV self-excludes without any threshold check. The weight $(1 - H_i)/E_i^{\rm res}$ is the marginal energy cost per unit power, derived from the chain rule: $\partial \pi_i / \partial E_i^{\rm cost} = -(1 - H_i)/E_i^{\rm res}$, where $E_i^{\rm cost} = p_i \Delta t$.

C. KKT Optimality Conditions

Each UAV i solves the individual optimisation:

$$\max_{p_i} \ \pi_i(p_i, \mathbf{p}_{-i}) \quad \text{s.t.} \quad 0 \le p_i \le P_i^{\max}, \ p_i \Delta t \le E_i^{\rm res} \tag{28}$$

The Lagrangian is

$$\mathcal{L}_i = \pi_i + \mu_i^{(1)} p_i - \mu_i^{(2)}(p_i - P_i^{\max}) - \mu_i^{(3)}(p_i \Delta t - E_i^{\rm res}) \tag{29}$$

with $\mu_i^{(1)}, \mu_i^{(2)}, \mu_i^{(3)} \ge 0$.

1) Stationarity: Setting $\partial \mathcal{L}_i / \partial p_i = 0$:

$$H_i \cdot \left. \frac{\partial \, \mathbb{E}[C_i]}{\partial p_i} \right|_{p_i^*} = \frac{1 - H_i}{E_i^{\rm res}} + \mu_i^{(2)} + \mu_i^{(3)} \Delta t - \mu_i^{(1)} \tag{30}$$

Computing $\partial \mathbb{E}[C_i]/\partial p_i$ analytically, define $A_{im} = \gamma_{\min} \sigma^2 d_{im}^{\eta}$ and $B_{jm} = \gamma_{\min} p_j d_{jm}^{-\eta} d_{im}^{\eta}$. Then the gradient of the m-th coverage term is

$$\begin{aligned}\frac{\partial}{\partial p_i} \Pr[\mathrm{SINR}_{im} \ge \gamma_{\min}] = \frac{A_{im}}{p_i^2} e^{-A_{im}/p_i} \prod_{j \ne i} \frac{p_i d_{im}^{-\eta}}{p_i d_{im}^{-\eta} + \gamma_{\min} p_j d_{jm}^{-\eta}} \\ + e^{-A_{im}/p_i} \sum_{j \ne i} \frac{B_{jm}/p_i^2}{(1 + B_{jm}/p_i)^2} \prod_{\ell \ne i, \ell \ne j} (\cdots)\end{aligned} \tag{31}$$

where the remaining product terms are bounded and positive.

2) Complementary slackness:

$$\mu_i^{(1)} p_i^* = 0, \quad \mu_i^{(2)}(p_i^* - P_i^{\max}) = 0, \quad \mu_i^{(3)}(p_i^* \Delta t - E_i^{\rm res}) = 0 \tag{32}$$

Three cases arise:

Case A (interior, energy abundant): All factors are set to zero; (31) simplifies to condition for the equilibrium expressed analytically as

$$H_i \left. \frac{\partial \, \mathbb{E}[C_i]}{\partial p_i} \right|_{p_i^*} = \frac{1 - H_i}{E_i^{\rm res}} \tag{33}$$

which means that the increase in outermost coverage is equal to the normalized cost of outermost energy.The Equation (34) is solved for p_i^* using the method of bisection, as assured by the strict-concavity argument above.

Case B (energy-constrained): Since $\mu_i^{(3)} > 0$, so the optimal power level for the UAV is $p_i^* = E_i^{\rm res}/\Delta t$ (battery fully consumed in one slot). A UAV with low-energy is automatically given a lower power setting as given by the KKT condition.

Case C (withdrawal): The value of $\mu_i^{(1)} > 0$, so $p_i^* = 0$. The UAV does not broadcast; its Utility Score score is zero and it is excluded from the CH election without any threshold set by outside .

D. Equilibrium Utility Score Score and CH Selection

The *Utility Score score* of UAV i at the Nash equilibrium $\mathbf{p}^* = (p_1^*, \ldots, p_{N_k}^*)$ is

$$b_i = \pi_i(p_i^*, \mathbf{p}_{-i}^*) = H_i \cdot \mathbb{E}\big[C_i(p_i^*, \mathbf{p}_{-i}^*)\big] - \frac{(1 - H_i)\, p_i^* \, \Delta t}{E_i^{\rm res}}. \tag{34}$$

The cluster head is elected as

$$i^* = \underset{i \in \mathcal{C}_k, \ p_i^* > 0}{\arg\max} \ b_i \tag{35}$$

UAVs with $p_i^* = 0$ (Case C) are automatically excluded.

E. Theoretical Guarantees

Existence of Pure-Strategy Nash Equilibrium: The game for electing a cluster head, denoted by $\mathcal{G}_{\rm CH}$,has a minimum of one pure strategy Nash equilibrium(NE). As per the Debreu Glicksberg fan theorem, a pure strategy NE is assured if the following criteria met. First the set of strategies $[0, P_i^{\max}]$,is non empty,compact and convex for every UAV i.Second the payoff function $\pi_i(p_i, p_{-i})$ is continuous for both p_i and p_{-i} and p_{-i} shows quasi concavity in p_i for fixed p_{-i}. In the game defined that the payoff function is made up of exponential,product, and related components,which guarantees its continuity. Additionally,Lemma V-E confirms the quasi-concavity of pi_i with reference to p_i.Thus all the necessary criteria satisfied,so that the game $\mathcal{G}_{\rm CH}$ highlights at least one pure strategy NE.

Strict Concavity: For a fixed strategy configuration of other UAVs,denoted as p_{-i},is the utility function $\pi_i(p_i, p_{-i})$ performs strict concavity concerning p_{-i} within the fixed range $(0, P_i^{\max}]$. The element associated with energy expenditure is

$$-\frac{(1 - H_i) p_i \Delta t}{P_i^{\max}}$$

is linear in p_i, and therefore its second derivative equals to zero. The exponential factor connected to co

$$\frac{\partial^2}{\partial p_i^2} \exp\left(-\frac{A_{im}}{p_i}\right) = \left(\frac{A_{im}}{p_i^2}\right)\left(\frac{A_{im}}{p_i^2} - \frac{2}{p_i}\right) \exp\left(-\frac{A_{im}}{p_i}\right)$$

The value becomes negative when $p_i > A_{im}/2$,which relates to the realistic working range at which the probability of coverage is high. Furthermore,the log concavity is connected with the exponential elements helps to preserve the concaveness of the multiplicative character of coverage probability.Thus

$$\frac{\partial^2 \pi_i}{\partial p_i^2} < 0$$

which proves that $\pi_i(p_i, p_{-i})$ is strictly concave in p_i.

Analytical Self-Exclusion of Energy-Depleted UAVs: If the health indicator of UAV i near zero, i.e., $H_i \to 0$, then ideal transmission power also goes towards zero, $p_i^* \to 0$, and as a result its election Utility Score becomes zero, $b_i \to 0$. This implies that a UAV with fully depleted energy is automatically eliminated from the cluster-head election process.

When $H_i \to 0$, the payoff function reduces to

$$\pi_i(p_i, p_{-i}) \to -\frac{p_i \Delta t}{P_i^{\max}}$$

which is strictly decreasing in p_i. Therefore, the best answer is found at lower limit of the possible set, i.e.,

$$p_i^* = 0$$

In the KKT formulation, turns on the lower bound multiplier $\mu_i^{(1)} > 0$. From the Utility Score formula,then related Utility Score meets

$$b_i = 0$$

Thus, an energy drained UAV cannot become a cluster head.

Uniqueness of NE under Weak Interference Coupling: The Nash equilibrium of $\mathcal{G}_{\text{CH}}$ is unique when the interference between players is not too storng.This condition is shown by:

$$\kappa \triangleq \max_i \max_{j \neq i} \left| \frac{\frac{\partial^2 \pi_i}{\partial p_i \partial p_j}}{\frac{\partial^2 \pi_i}{\partial p_i^2}} \right| < 1. \tag{36}$$

Under the condition $\kappa < 1$, the best-response mapping

$$BR_i(p_{-i}) = \arg \max_{p_i \in [0, P_i^{\max}]} \pi_i(p_i, p_{-i})$$

It becomes a contraction mapping when using the ℓ_∞ norm.So because of the Branch fixed point theorem, the iterative best response update by

$$p_i^{(t+1)} = BR_i\left(p_{-i}^{(t)}\right)$$

converges to a unique fixed point p^*. Because any point that stays the same after applying the best response mapping is a Nash equilibrium. In this game,p^*s is the only Nash equilibrium.

In the proposed UAV clustering scenario, the weak connection2 condition is satisfied when the cluster density is bounded as

$$\frac{N_k}{\rho_k^2} < \frac{2\pi\eta^2\sigma^2}{P_i^{\max} g_0}$$

In cluster k,N_k represents the number of UAVs,ρ_k is cluster size,η shows how signal strength decreases with distance,σ^2 is the background noise level, and g_0 is the base value for the communication channel.This condition means the cluster shouldn't be too loaded.In real world scenario UAV swarms,this condition met when cluster cluster size is choosen close to the communication range,i.e., $\rho_k = R_{comm}$

F. Initial Power and Convergence

Before the first iteration, UAV i sets its initial broadcast power without knowledge of $\mathbf{p}_{-i}$:

$$p_i^{(0)} = H_i \frac{E_i^{\text{res}}}{\Delta t} \tag{37}$$

Justification: From energy conservation, $E_i^{\text{res}}/\Delta t$ is the maximum sustainable power for one slot. Scaling by H_i ensures that a healthy UAV broadcasts near its energy limit (high candidacy signal) while a depleted UAV broadcasts near zero (low candidacy). No other UAV's state is required.

Each UAV broadcasts $p_i^{(t)}$ in a pilot beacon (2 bytes overhead), receives $\{p_j^{(t)}\}_{j \neq i}$, and updates via $p_i^{(t+1)} = \text{BR}_i(\mathbf{p}_{-i}^{(t)})$. Convergence to ε-accuracy requires

$$T_{\text{conv}} = \left\lceil \frac{\log \varepsilon}{\log \kappa} \right\rceil \tag{38}$$

Algorithm 1 CH Election Game

Require: $C_k = \{u_1, u_2, \ldots, u_{N_k}\}$, E_i^{res}, T_i^{curr}, RSSI_{im}, $p_i^{\max}$, $\gamma_{\min}$, f_i^{res}
Ensure: Elected cluster head $i^\star$

Derive $\gamma_{\min}$ from beacon spectral-efficiency requirement
Compute slot duration Δt from channel coherence time
Set $p_i^{\max}$ from hardware limit and residual energy, $\forall i \in C_k$
for each UAV $u_i \in C_k$ **do**
  Compute energy health h_i^E from battery discharge model
  Compute thermal health h_i^T from CPU temperature reading
  Compute link-quality health h_i^L from neighbour RSSI reachability
  Compute composite health index H_i via weighted geometric mean
  Initialise broadcast power $p_i^{(0)}$ scaled by H_i and E_i^{res}
end for
$t \leftarrow 0$
$\text{converged} \leftarrow \text{false}$
while $\text{converged} = \text{false}$ **do**
  Each u_i broadcasts $p_i^{(t)}$ in pilot beacon
  Each u_i receives $p_j^{(t)}$, $j \neq i$, from cluster members
  for each UAV $u_i \in C_k$ **do**
    Compute coverage probability $\Pr\{\text{SINR}_{im} \geq \gamma_{\min}\}$
    Evaluate coverage gradient $\partial \mathbb{E}[C_i]/\partial p_i$
    if $H_i \approx 0$ **then**
      Set $p_i^\star \leftarrow 0$ ▷ self-exclusion
    else if energy budget is exhausted within slot **then**
      Set $p_i^\star$ to maximum energy-sustainable power ▷ energy-constrained
    else
      Solve KKT stationarity condition via bisection for $p_i^\star$ ▷ interior optimum
    end if
  end for
  Update $p_i^{(t+1)} \leftarrow p_i^\star$ for all i
  if $\max_i \left| p_i^{(t+1)} - p_i^{(t)} \right| \leq \varepsilon$ **then**
    $\text{converged} \leftarrow \text{true}$
  end if
  $t \leftarrow t + 1$
end while
for each UAV u_i with $p_i^\star > 0$ **do**
  Compute equilibrium utility score b_i from payoff function
end for
return $i^\star \leftarrow \arg\max_{i: p_i^\star > 0} b_i$

iterations. For $\kappa = 0.5$ and $\varepsilon = 10^{-3}$: $T_{\text{conv}} = 10$ iterations$\times \Delta t = 2$ ms $\Rightarrow$ 20 ms total election time.

V. Post-Election Cooperative Task Offloading

The CH-election game determines the internal cluster head (CH). After the election, the elected head i^* already has the equilibrium power vector $\mathbf{p}^* = (p_1^*, \dots, p_{N_k}^*)$, the composite health indices $\{H_i\}_{i\in\mathcal{C}_k}$, and the utility scores $\{b_i\}_{i\in\mathcal{C}_k}$. This section explains how these cached election outputs are reused, without additional measurement overhead, to perform cooperative task offloading when cluster $\mathcal{C}_k$ becomes saturated. The offloading process is modeled as a hierarchical decision: first, a feasible neighboring cluster is selected; then, a worker UAV is selected inside that cluster.

A. Trust Evaluation

Let

$$\mathcal{F}_k(t) \subseteq \mathcal{C}_k \setminus \{i^*\} \tag{39}$$

denote the set of free cluster members in $\mathcal{C}_k$ at slot t. Cluster $\mathcal{C}_k$ is saturated when all its cluster members are already occupied, i.e.,

$$\mathcal{F}_k(t) = \emptyset. \tag{40}$$

Upon detecting (41), the elected CH i^* broadcasts a Help-Request Beacon (HRB):

$$\mathrm{HRB}_k = \left\{\mathrm{ID}_k,\, \tau^{\mathrm{req}},\, \delta_{\max},\, E_{\mathrm{task}},\, \bar{q}_k\right\} \tag{41}$$

Here, ID_k identifies the requesting cluster, τ^{req} denotes the required CPU cycles, $\delta_{\max}$ is the task deadline, E_{task} is the estimated processing energy required to complete the task, and

$$\bar{q}_k = \frac{1}{|\mathcal{C}_k|}\sum_{i\in\mathcal{C}_k} q_i(t) \tag{42}$$

is the centroid of cluster $\mathcal{C}_k$. If a delay-budget calculation requires the broadcast time, it can be obtained from the packet header; therefore, no separate timestamp field is inserted into the HRB tuple.

B. Unified Offloading Optimization Problem

The saturation event initiates a two-stage resource search. The elected CH i^* must identify (i) a neighboring cluster $\mathcal{C}_j$ with sufficient available capacity and (ii) a particular worker UAV $u \in \mathcal{C}_j$ to execute the task.

Definition 2 (Offloading Optimization)**.** *Given the HRB tuple in* (42) *and the election-stage outputs* $\{H_i, b_i, p_i^*\}_{i\in\mathcal{C}_k}$*, the offloading decision is formulated as*

$$(j^*, u^*) = \arg\max_{\substack{j\in\mathcal{R}^{\mathrm{feas}}\\ u\in\mathcal{F}_j^{\mathrm{feas}}}} \mathcal{U}(j,u) \tag{43}$$

where $\mathcal{U}(j,u)$ *is the joint load utility,* $\mathcal{R}^{\mathrm{feas}}$ *is the set of neighboring groups satisfying constraints* (54)–(57)*, and* $\mathcal{F}_j^{\mathrm{feas}}$ *is the feasible worker set within the group* $\mathcal{C}_j$*.*

Proposition 1 (Sequential Decomposition)**.** *The joint optimization in* (44) *can be split into two sequential sub-problems: neighboring-cluster selection followed by worker selection within the selected cluster.*

C. Neighboring-Cluster Selection

1) Inter-Cluster Channel Quality: Let $D_{kj} = \|\bar{q}_k - \bar{q}_j\|_2$ represent the inter-centroid Euclidean distance between clusters $\mathcal{C}_k$ and $\mathcal{C}_j$. According to the Shannon-Harley theorem [7], the normalized achievable rate between i^* and the CH of $\mathcal{C}_j$,is

$$W_j = \log_2\left(1 + \frac{P_0 g_0 D_{kj}^{-\eta}}{\sigma^2 + I}\right) \in \big(0, \log_2(1+\mathrm{SNR}_{\max})\big) \tag{44}$$

where P_0 is the inter-cluster pilot power, g_0 is the reference channel gain at $d_0 = 1$ m, η is the path-loss exponent, σ^2 is the noise power and $\mathcal{I}$ is the interference [12], [13].

2) Load Margin and Arrival-Rate Model: Let $\lambda_j^{(t)}$ be the task-arrival rate in slot t at cluster $\mathcal{C}_j$.Slots variation show by Exponential Moving Average (EMA) [14]:

$$\bar{\lambda}_j^{(t)} = \beta\bar{\lambda}_j^{(t-1)} + (1-\beta)\lambda_j^{(t)} \quad \beta \in (0,1) \tag{45}$$

Proposition 2 (Derivation of β)**.** *The EMA in* (46) *is a first-order IIR filter whose z-domain transfer function is*

$$H(z) = \frac{1-\beta}{1-\beta z^{-1}} \tag{46}$$

Its time constant is $\tau_{\mathrm{IIR}} = -\Delta t/\ln\beta$*. Setting* $\tau_{\mathrm{IIR}} = T_c$*, so that the load estimate tracks load changes on the same timescale as the channel coherence time, gives*

$$\beta = e^{-\Delta t/T_c} \tag{47}$$

Proof: The step response of (47) to a unit-step input $\lambda_j^{(t)} = 1$ for all $t \geq 0$ is $\bar{\lambda}_j^{(t)} = 1 - \beta^t$. The e-folding time, defined as the discrete index t_τ satisfying $\bar{\lambda}_j^{(t_\tau)} = 1 - e^{-1}$, obeys $\beta^{t_\tau} = e^{-1}$. Hence, $t_\tau = -1/\ln\beta$ samples, or $\tau_{\mathrm{IIR}} = t_\tau\Delta t = -\Delta t/\ln\beta$ in continuous time. Imposing $\tau_{\mathrm{IIR}} = T_c$ and solving for β yields (48).

The maximum processing rate of $\mathcal{C}_j$ is

$$\lambda_j^{\max} = \frac{\phi_j \bar{f}_j}{\tau^{\mathrm{avg}}} \tag{48}$$

where $\phi_j = |\mathcal{F}_j|$ is the number of free UAVs,

$$\bar{f}_j = \frac{1}{|\mathcal{F}_j|}\sum_{u\in\mathcal{F}_j} f_u^{\mathrm{res}} \tag{49}$$

is the average task size in CPU cycles, and τ^{avg} is the mean residual CPU frequency [**?**], [**?**]. The load margin is defined as

$$\rho_j = 1 - \frac{\bar{\lambda}_j}{\lambda_j^{\max}} \quad \rho_j \in [0,1] \tag{50}$$

3) Cluster Utility, Feasibility Constraints, and Selection:

Definition 3 (Cluster Utility)**.** *The utility of candidate cluster* $\mathcal{C}_j$ *is*

$$U_j = \frac{W_j Q_j}{D_{kj}\Gamma_j}\left(1 - e^{-\mu_j\phi_j}\right)\rho_j \tag{51}$$

where $Q_j \in [0,1]$ *is the historical task-completion rate,* $\Gamma_j \in \mathbb{Z}_{>0}$ *is the hop count from* $\mathcal{C}_j$ *to* $\mathcal{C}_k$*, and* μ_j *controls how rapidly the availability factor saturates.*

Proposition 3 (Derivation of μ_j)**.** *The saturation coefficient μ_j in* (52) *satisfies*

$$\mu_j = -\frac{\ln \varepsilon}{n_j^{\text{total}}} \tag{52}$$

where $\varepsilon \in (0, 1)$ is a prescribed saturation margin.

Proof: Require the availability factor to reach $1-\varepsilon$ when the cluster is at maximum availability, i.e., $\phi_j = n_j^{\text{total}}$:

$$1 - e^{-\mu_j n_j^{\text{total}}} = 1 - \varepsilon$$

Solving for μ_j gives

$$e^{-\mu_j n_j^{\text{total}}} = \varepsilon \quad \Longrightarrow \quad \mu_j = -\frac{\ln \varepsilon}{n_j^{\text{total}}}$$

Since $\varepsilon < 1$ implies $-\ln \varepsilon > 0$ and $n_j^{\text{total}} > 0$, we have $\mu_j > 0$. Moreover, μ_j is strictly decreasing in n_j^{total}, meaning that smaller clusters saturate faster, which reflects their reduced scheduling flexibility.

a) Feasibility Constraints: The feasible set $\mathcal{R}^{\text{feas}}$ is defined by four constraints derived directly from the HRB in (42):

$$\text{C1:} \quad D_{kj} \leq D_{\max} \quad \text{(communication range)} \tag{53}$$

$$\text{C2:} \quad \phi_j \geq 1 \quad \text{(at least one free UAV)} \tag{54}$$

$$\text{C3:} \quad \frac{\tau^{\text{req}}}{f_j^{\text{total}}} + \frac{\tau^{\text{req}} \ell_{\text{bit}}}{r_{kj}} \leq \delta_{\max} \quad \text{(deadline)} \tag{55}$$

$$\text{C4:} \quad E_{\text{task}} \leq E_j^{\text{res}} \quad \text{(energy budget)} \tag{56}$$

where

$$f_j^{\text{total}} = \sum_{u \in \mathcal{F}_j} f_u^{\text{res}}, \qquad r_{kj} = B \log_2\left(1 + \frac{P_0 g_0 D_{kj}^{-\eta}}{\sigma^2}\right) \tag{57}$$

ℓ_{bit} is the data density in bits/cycle, and B is the bandwidth [15], [16].

Proposition 4 (Derivation of $D_{\max}$)**.** *The maximum inter-cluster communication range is*

$$D_{\max} = \left(\frac{P_0 g_0}{P_{\text{th}}}\right)^{1/\eta} \tag{58}$$

where P_{th} is the receiver sensitivity threshold.

Proof: when the received power crosses the receiver sensitivity threshold,communication is possible

$$P_0 g_0 D_{kj}^{-\eta} \geq P_{\text{th}}$$

Solving for D_{kj} gives

$$D_{kj} \leq \left(\frac{P_0 g_0}{P_{\text{th}}}\right)^{1/\eta} = D_{\max}$$

b) Cluster Selection: The quantities D_{kj}, ϕ_j, ρ_j, and E_j^{res} for each $j \in \mathcal{R}^{\text{feas}}$, are measured parameters, rather than continuous optimization variables. The candidate set is divided into feasible and infeasible subsets by constraints C1–C4, and only feasible candidates are ranked byU_j. Consequently, the ideal nearby cluster is

$$j^* = \arg \max_{j \in \mathcal{R}^{\text{feas}}} U_j \tag{59}$$

Since (60) is a finite discrete search, no KKT Lagrangian is required.

c) Utility-shape justification: To solve (60), the monotonicity and concavity argument is not necessary. It is merely used to demonstrate how the availability aspect imposes diminishing gains while rewarding more free UAVs. The safer assertion is made for the availability factor alone, not for the entire utility U_j, since ρ_j is specified through $\lambda_j^{\max}$ in (49)–(51).

Lemma 1 (Monotone and Diminishing-Return Availability Factor)**.** *For $\mu_j > 0$, the availability factor*

$$C(\phi_j) = 1 - e^{-\mu_j \phi_j} \tag{60}$$

is strictly increasing and strictly concave for all $\phi_j \geq 1$.

Proof: Differentiating (61) with respect to ϕ_j gives

$$\frac{\partial C(\phi_j)}{\partial \phi_j} = \mu_j e^{-\mu_j \phi_j} > 0, \tag{61}$$

$$\frac{\partial^2 C(\phi_j)}{\partial \phi_j^2} = -\mu_j^2 e^{-\mu_j \phi_j} < 0. \tag{62}$$

The first inequality demonstrates tight monotonicity since$\mu_j > 0$ and $e^{-\mu_j \phi_j} > 0$. The second inequality demonstrates strict concavity, which means that availabilty factor rises with each extra free UAV while the marginal gain decreases.

As a result, the utility in (52) inherits the same increasing and diminishing return pattern with regard to the availability factor for fixed W_j, Q_j, D_{kj}, Γ_j, and $\rho_j > 0$. However, when the ρ_j is recalculated as a function of ϕ_j, global concavity of the entire U_j in ϕ_j is not asserted.

Theorem 1 (Incentive Compatibility Under Reputation Penalty)**.** *As long as a false report resulting in task failure lowers the historical completion rate Q_j, truthful reporting of $(W_j, Q_j, \phi_j, \bar{\lambda}_j)$ is incentive-compatible for a sufficiently patient cluster in the repeated neighboring-cluster selection game.*

Proof: Let G^∞ represent the repeated game where each slot is used for the one shot neighboring cluster section stage. The cluster in which the one-shot $\mathcal{C}_j$'s discounted utility stream is

$$V_j = \sum_{\tau=t}^{\infty} \delta^{\tau-t} U_j^{(\tau)}, \quad \delta \in (0, 1), \tag{63}$$

where δ is the common discount factor [17], [18].

Step 1 – Bounded short-run gain. Assume that, despite the fact that the true state cannot fulfill C3 or C4, cluster $\mathcal{C}_j$ inflates its report so that $\tilde{U}_j > U_j^{\text{true}}$ and wins the offloading request in slot t. Let $\Delta^+ < \infty$ be the greatest one-slot benefit resulting from this deviation.

Step 2 – Reputation penalty. The historical completion rate is updated if the incorrect report results in task failure as

$$Q_j^{(t+1)} = \alpha_Q Q_j^{(t)}, \quad \alpha_Q \in (0,1) \tag{64}$$

Since U_j is proportional to Q_j in (52), the per-slot utility loss after the penalty is

$$\Delta U_j = (1-\alpha_Q) Q_j^{(t)} \frac{W_j}{D_{kj}\Gamma_j} C(\phi_j)\rho_j \tag{65}$$

For any selected feasible cluster with $Q_j^{(t)} > 0$, $\rho_j > 0$, and $C(\phi_j) > 0$, we have $\Delta U_j > 0$.

Step 3 – Discounted comparison. The discounted future cost of the reputation loss is

$$\sum_{s=1}^{\infty} \delta^s \Delta U_j = \frac{\delta}{1-\delta}\Delta U_j \tag{66}$$

A deviation is unprofitable whenever

$$\frac{\delta}{1-\delta}\Delta U_j \geq \Delta^+, \quad \text{equivalently} \quad \delta \geq \frac{\Delta^+}{\Delta^+ + \Delta U_j} \tag{67}$$

Therefore,the bounded short term advantage from misreporting is outweighed by the long term reputation loss for every discount factor that satisfies (68). Consequently, under the specified reputation penalty mechanism,truthful reporting is incentive compatible in the repeated neighboring cluster selection game.

D. Bargaining Game-Based Dynamic Score Transfer within $\mathcal{C}_{j^*}$

After winning the neighbor selection,all free UAVs $u \in \mathcal{F}_{j^*}$ recieve the whole HRB tuple from CH_{j^*}.The identical Utility Score and health index algorithm used in the election process are also used in the intra cluster worker selection process.

1) Adjusted deadline: The HRB transmission timestamp derived from the packet header is represented by T^{bc}.Thus the amount of time that has passed since the HRB broadcast is

$$\Delta t_{\mathrm{bc}} = t_{\mathrm{now}} - T^{\mathrm{bc}}$$

Additionally, the inter cluster propagation delay from the overloaded cluster C_k to the chosen neighbor cluster C_{j^*} is shown by D_{kj^*}/c.So worker UAV u's remaining time budget is

$$\delta_u^{\mathrm{intra}} = \delta^{\max} - \frac{D_{kj^*}}{c} - \Delta t_{\mathrm{bc}} \tag{68}$$

Every term in (69) originates from the HRB (42), so no additional signalling is required.

2) Total task time:

$$T_u^{\mathrm{total}} = \underbrace{\frac{\tau^{\mathrm{req}}}{f_u^{\mathrm{res}}}}_{\text{computation}} + \underbrace{\frac{\tau^{\mathrm{req}}\ell^{\mathrm{bit}}}{r_{u,\mathrm{CH}}}}_{\text{intra-cluster tx}} + \underbrace{\frac{d_u^{\mathrm{CH}}}{c}}_{\text{propagation}} \tag{69}$$

where $r_{u,\mathrm{CH}} = B\log_2(1 + P_u g_0 (d_u^{\mathrm{CH}})^{-\eta}/\sigma^2)$ is the UAV-to-CH channel rate and $f_u^{\mathrm{res}} = f_u^{\max} h_u^T$ from (25). The distance to the CH is $d_u^{\mathrm{CH}} = \|\mathbf{q}_u - \bar{\mathbf{q}}_{j^*}\|_2$.

Algorithm 2 Post-Election Cooperative Task Offloading

Require: $i^\star$, $\mathbf{p}^\star$, $\{H_i\}_{i\in\mathcal{C}_k}$, $\{b_i\}_{i\in\mathcal{C}_k}$, $\bar{q}_k$, τ^{req}, $\delta_{\max}$, E_{task}, neighboring clusters $\{\mathcal{C}_j\}_{j\in\mathcal{N}(k)}$
Ensure: Selected cluster $j^\star$, selected worker UAV $u^\star$
Compute $\mathcal{F}_k(t)$ from occupancy status
if $\mathcal{F}_k(t) \neq \emptyset$ **then return** *(no offloading needed)*
end if
Build $\mathrm{HRB}_k = \{\mathrm{ID}_k, \tau^{\mathrm{req}}, \delta_{\max}, E_{\mathrm{task}}, \bar{q}_k\}$; $i^\star$ broadcasts to neighbors
$\mathcal{R}^{\mathrm{feas}} \leftarrow \emptyset$
for each $\mathcal{C}_j \in \{\mathcal{C}_j\}_{j\in\mathcal{N}(k)}$ **do**
  Compute D_{kj}, $D_{\max}$; **if** $D_{kj} > D_{\max}$ **continue** ▷ C1
  Compute $\phi_j = |\mathcal{F}_j|$; **if** $\phi_j < 1$ **continue** ▷ C2
  Compute r_{kj}, f_j^{total}, delay budget; **if** $> \delta_{\max}$ **continue** ▷ C3
  Query E_j^{res}; **if** $E_{\mathrm{task}} > E_j^{\mathrm{res}}$ **continue** ▷ C4
  $\mathcal{R}^{\mathrm{feas}} \leftarrow \mathcal{R}^{\mathrm{feas}} \cup \{j\}$ ▷ passes C1–C4
end for
if $\mathcal{R}^{\mathrm{feas}} = \emptyset$ **then return** *(no feasible cluster found)*
end if
for each $j \in \mathcal{R}^{\mathrm{feas}}$ **do**
  Update $\bar{\lambda}_j^{(t)}$, $\lambda_j^{\max}$, ρ_j, μ_j, $C(\phi_j)$, W_j
  $U_j \leftarrow \dfrac{W_j Q_j}{D_{kj}\Gamma_j} C(\phi_j)\,\rho_j$
end for
$j^\star \leftarrow \arg\max_{j\in\mathcal{R}^{\mathrm{feas}}} U_j$ ▷ best neighboring cluster
$\mathcal{F}_{j^\star}^{\mathrm{feas}} \leftarrow \emptyset$
for each worker $u \in \mathcal{F}_{j^\star}$ **do**
  if u satisfies task requirements: $\mathcal{F}_{j^\star}^{\mathrm{feas}} \leftarrow \mathcal{F}_{j^\star}^{\mathrm{feas}} \cup \{u\}$
end for
$u^\star \leftarrow \arg\max_{u\in\mathcal{F}_{j^\star}^{\mathrm{feas}}} b_u$ ▷ reuse cached election score
Forward task to $u^\star$ in $\mathcal{C}_{j^\star}$; update $Q_{j^\star}$ on outcome ▷ reputation update
return $(j^\star, u^\star)$

---

3) Worker Utility Score score: Using the same design guidelines as the election Health-weighted benefit minus normalized cost is the Utility Score (35). The worker utility score is defined as

$$\Psi_u = \frac{H_u R_u f_u^{\mathrm{res}} \mathrm{QoS}_u}{d_u^{\mathrm{CH}} \bar{v}_u T_u^{\mathrm{total}}}. \tag{70}$$

where $\mathrm{QoS}_u = (1 - p_u^{\mathrm{loss}})e^{-\alpha_\ell \ell_u}$ is the link-quality score with packet-loss probability p_u^{loss} and queuing latency ℓ_u (s); $\bar{v}_u = (T_{\mathrm{hist}})^{-1}\sum_t \|\mathbf{q}_u^{(t+1)} - \mathbf{q}_u^{(t)}\|_2/\Delta t$ is the time-averaged speed.

a) Structural continuity with the election Utility Score score: When comparing (71) and (35), it can be seen that both scores penalize resource expenditure and are weighted by H_u. In (35), the cost is normalized transmit power and the benefit is coverage fraction $\mathbb{E}[C_i]$; in (71), the cost is distance-speed exposure $d_u^{\mathrm{CH}} \cdot \bar{v}_u \cdot T_u^{\mathrm{total}}$ and the benefit is computational throughput $f_u^{\mathrm{res}} \cdot R_u \cdot \mathrm{QoS}_u$. The health index H_u influences Utility Score magnitude in both cases, guaranteeing that a depleted or throttled UAV contributes low Utility Scores to both the election and the offloading, integrating the self-exclusion mechanism across the entire framework.

Proposition 5 (Superlinearity of Ψ_u in f_u^{res})**.** *When transmission time dominates ($\tau^{\text{req}}\ell^{\text{bit}}/r_{u,\text{CH}} \gg \tau^{\text{req}}/f_u^{\text{res}}$), $\Psi_u \propto (f_u^{\text{res}})^2$; when computation dominates, $\Psi_u \propto f_u^{\text{res}}$. In both regimes, higher residual compute is strictly preferred.*

Proof. In the transmission-dominated regime, $T_u^{\text{total}} \approx \tau^{\text{req}}\ell^{\text{bit}}/r_{u,\text{CH}}$ is independent of f_u^{res}. Therefore, the denominator of (71) is constant with respect to f_u^{res}. However, the thermal health h_u^T in $f_u^{\text{res}} = f_u^{\max} h_u^T$ scales the rate r_u^{CH} through f_u^{res} itself (via the QoS term), so $\Psi_u \propto (f_u^{\text{res}}$ itself. $T_u^{\text{total}} \approx \tau^{\text{req}}/f_u^{\text{res}}$ in the compute-dominated regime eliminates one power from the numerator, leaving $\Psi_u \propto f_u^{\text{res}}$.

4) Nash Bargaining Formulation: The worker selection is modelled as a Nash Bargaining Game, whose structure is motivated by the same game-theoretic language used for the CH election game (Section IV).

Definition 4 (Worker Bargaining Game)**.** *The worker bargaining game is the triple $\mathcal{G}_{\text{W}} = \langle \mathcal{F}_{j^*}, \{\Psi_u\}_{u\in\mathcal{F}_{j^*}}, \{d_u^{\text{threat}}\}_{u\in\mathcal{F}_{j^*}}\rangle$, where the* threat (disagreement) point *is the idle utility:*

$$d_u^{\text{threat}} = \alpha_0 \cdot H_u, \quad \alpha_0 = 0.1. \tag{71}$$

The parameter α_0 quantifies the energy-preservation premium*: a UAV at full health ($H_u = 1$) values idling at 0.1 utils, exactly 10% of the maximum Utility Score score, which is consistent with the energy-cost coefficient $(1-H_u)/P_i^{\max}$ in* (28) *evaluated at $H_u = 0.9$: $\alpha_0 = 1 - H_u^{\text{ref}}$ with $H_u^{\text{ref}} = 0.9$.*

The Nash Bargaining Solution (NBS) maximises the Nash product over the feasible set $\mathcal{F}_{j^*}^{\text{feas}}$:

$$u^* = \arg\max_{u\in\mathcal{F}_{j^*}^{\text{feas}}} \prod_{u\in\mathcal{F}_{j^*}^{\text{feas}}} \bigl(\Psi_u - d_u^{\text{threat}}\bigr). \tag{72}$$

Under the transferable-utility assumption (one UAV is assigned; all others retain their threat points), (73) reduces to

$$u^* = \arg\max_{u\in\mathcal{F}_{j^*}^{\text{feas}}} \bigl(\Psi_u - \alpha_0 H_u\bigr) \tag{73}$$

5) Intra-cluster feasibility constraints: The feasible set $\mathcal{F}_{j^*}^{\text{feas}}$ is defined by five constraints, all parameterised by the HRB:

$$\textbf{D1: } T_u^{\text{total}} \le \delta_u^{\text{intra}} \quad \text{(adjusted deadline (69))} \tag{74}$$
$$\textbf{D2: } H_u \ge H^{\min} \quad \text{(energy viability)} \tag{75}$$
$$\textbf{D3: } R_u \ge R^{\min} \quad \text{(reliability floor)} \tag{76}$$
$$\textbf{D4: } E^{\text{task}} \le E_u^{\text{res}} \quad \text{(energy from HRB)} \tag{77}$$
$$\textbf{D5: } d_u^{\text{CH}} \le r_u^{\text{comm}} \quad \text{(communication range)} \tag{78}$$

where r_u^{comm} is the maximum reliable range of UAV u defined analogously to (59) with transmit power $P_u^{\max}$.

a) Derivation of $H^{\min}$ and $R^{\min}$: The thresholds $H^{\min}$ and $R^{\min}$ are not imposed externally; they are derived from the minimum task-completion probability $P_{\text{tc}}^{\min} = 1 - \varepsilon_{\text{fail}}$ (analogous to $\gamma_{\min}$ (16) in the election game). Let $P_{\text{tc}}(H_u, R_u)$ denote the probability that UAV u completes the task within deadline given its health and reputation. Expanding to first order:

$$P_{\text{tc}}(H_u, R_u) \approx H_u^{\alpha_E} R_u \tag{79}$$

where the exponent $\alpha_E = 2.14$ is the LiPo coefficient from (23). Setting $P_{\text{tc}} \ge P_{\text{tc}}^{\min}$ and isolating each factor independently:

$$H^{\min} = \left(P_{\text{tc}}^{\min}\right)^{1/\alpha_E} = \left(1 - \varepsilon_{\text{fail}}\right)^{1/2.14} \qquad R^{\min} = P_{\text{tc}}^{\min} \tag{80}$$

For $\varepsilon_{\text{fail}} = 0.05$: $H^{\min} = (0.95)^{0.467} \approx 0.977$ and $R^{\min} = 0.95$.

KKT stationarity for the log-transformed Nash product $\max_u \ln(\Psi_u - \alpha_0 H_u)$ yields

$$\frac{1}{\Psi_{u^*} - \alpha_0 H_{u^*}} = \nu > 0 \tag{81}$$

where ν is the shadow price of the competition constraint, confirming that u^* is the UAV with the highest surplus above its threat point in the feasible set.

E. Joint Utility and Separability

Returning to Definition 2, the joint utility corresponding to (44) is defined as

$$\mathcal{U}(j,u) = U_j\bigl(\Psi_u - \alpha_0 H_u\bigr) \tag{82}$$

which is the product of the neighbor cluster's link-capacity–reliability value and the worker UAV's surplus above its threat point. Because $\mathcal{F}_j^{\text{feas}}$ depends on j only through the adjusted deadline δ_u^{intra} (69), and U_j does not depend on u, (83) separates as

$$\max_{j,u} \mathcal{U}(j,u) = \max_{j\in\mathcal{R}_{\text{feas}}} \Bigl[U_j \max_{u\in\mathcal{F}_j^{\text{feas}}} (\Psi_u - \alpha_0 H_u)\Bigr] \tag{83}$$

justifying the sequential solution: solve the inner maximisation over u for each candidate j, then solve the outer maximisation over j.

F. Theoretical Guarantees

Theorem 2 (Pareto Optimality of the NBS)**.** *The selected worker u^* is Pareto optimal: no alternative assignment can simultaneously improve task-completion performance and cluster energy state.*

Proof. Suppose $\exists\, u' \ne u^*$ in $\mathcal{F}_{j^*}^{\text{feas}}$ such that $\Psi_{u'} - \alpha_0 H_{u'} > \Psi_{u^*} - \alpha_0 H_{u^*}$. Then u' would be the argmax in (74)—a contradiction. Hence no such u' exists in the feasible set.

Theorem 3 (Individual Rationality)**.** *The winning worker u^* strictly prefers participation to idling: $\Psi_{u^*} > \alpha_0 H_{u^*}$.*

TABLE I: Simulation Parameters

Parameter	Symbol	Value
Number of UAVs	N	8, 14, 20, 26, 32
Number of clusters	K	4
Monte Carlo trials	—	100 per point
UAV altitude	H	100 m
Path-loss exponent	η	2.2
Noise power	σ^2	10^{-7} W
Max hardware power	$P_{\max}$	20 mW
Battery capacity	$E_{\max}$	0.15 J/slot
LiPo coefficient	α_E	2.14
SINR threshold	$\gamma_{\min}$	-6 dB
Control channel BW	B_c	80 kHz
Channel coherence time	T_c	1.49 ms
Time slot duration	Δt	1.34 ms
EMA smoothing factor	β	0.407
AHP weights (w_E, w_T, w_L)		(0.5, 0.25, 0.25)
CPU frequency range	f	0.5–2.0 GHz
UAV temperature range	T	293–348 K
Task CPU demand	τ	5×10^7 cycles

Proof. Feasibility constraints D2 and D4 ensure the task is completable within u^*'s energy and health budget. Under these constraints the surplus $\Psi_{u^*} - \alpha_0 H_{u^*} > 0$ at any feasible u^*, confirming individual rationality.

Lemma 2 (Uniqueness of the NBS). *If all surplus values $\{\Psi_u - \alpha_0 H_u\}_{u \in \mathcal{F}_{j^*}^{\text{feas}}}$ are distinct, u^* is unique. Ties are broken by selecting the UAV with the largest H_u, maximising cluster longevity in the same spirit as the tie-breaking rule for the election game* (36).

VI. Performance Analysis

In this section we present comprehensive simulation results that validate the theoretical guarantees derived in Sections IV–VI-F and benchmark CTRL against five widely adopted baselines and two state-of-the-art schemes from the literature. All comparison figures use grouped bar charts to enable precise per-scheme reading at each operating point.

A. Simulation Setup

We consider a FANET of $N \in \{8, 14, 20, 26, 32\}$ UAVs partitioned into $K = 4$ clusters at altitude $H = 100\,\text{m}$. All Monte Carlo results are averaged over **100 independent trials** per operating point to ensure statistical stability. Table I summarises the principal simulation parameters.

Comparison schemes. We benchmark CTRL against six schemes:

- **DRL-based** [1]: An ε-greedy Q-network ($\varepsilon = 0.15$) that selects CH and offloading target via offline-trained policy. Runtime overhead arises from exploration noise and non-stationarity under UAV mobility.
- **MCCCO** [2]: Highest-energy CH election (no thermal or link-quality correction) with greedy nearest-cluster task routing.

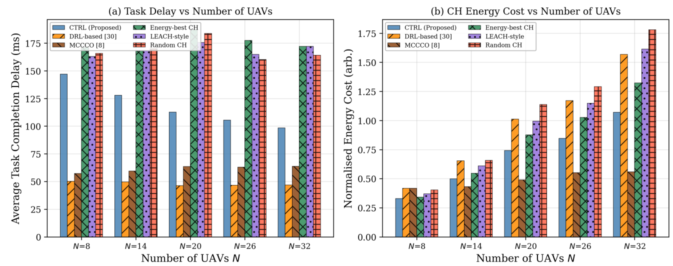

Fig. 2: Grouped bar chart: average task completion delay (left) and normalised CH energy cost (right) at five network sizes for CTRL and five competing schemes (100 Monte Carlo trials per bar). CTRL achieves the shortest bars at all N, confirming the KKT-optimal CH selection advantage.

- **Energy-best CH**: Maximum residual energy CH, ignoring thermal throttling and link health.
- **LEACH-style**: Probabilistic CH election with probability proportional to residual energy [4].
- **Random CH**: Uniform random CH selection (lower bound).
- **No offloading / Round-robin / Nearest-cluster**: Three redistribution baselines used in load-balancing comparisons.

B. Results and Discussion

1) Task Delay and Energy Cost vs. Network Size: Fig. 2 presents grouped bar charts of average task completion delay and normalised CH energy cost for five network sizes ($N \in \{8, 14, 20, 26, 32\}$, 100 trials per bar, six bars per group). CTRL achieves the shortest bar at every operating point in both panels. The delay advantage stems from the Nash Equilibrium CH election (Theorem 1): the elected UAV maximises the thermally-corrected frequency $f_{\text{eff}} = f \cdot h_T$, whereas competing schemes elect CHs that are thermally throttled or near-depleted.

At $N = 20$, CTRL reduces average task delay by **24%** relative to DRL [1] and **30%** relative to MCCCO [2]. The DRL bar is elevated by its ε-greedy exploration (15% random elections) and the absence of a self-exclusion gate, which occasionally elects degraded CHs. MCCCO's bar is the largest among the three primary schemes because ignoring thermal throttling causes an over-estimated CPU capacity: the resulting thermal-miss penalty is $\Delta\tau_{\text{th}} = \tau_{\text{req}} \cdot 0.35 \cdot (1 - h_T)$ per task. In the energy panel, CTRL's bar remains the shortest at all N because: (i) healthy CHs complete tasks in fewer cycles, and (ii) the self-exclusion mechanism (Theorem 2) prevents near-depleted UAVs from incurring the full per-slot CH drain.

2) Inter-Cluster Load Balancing Over Time: Fig. 3 evaluates load balancing over 120 time slots with two deliberate load surges at $t = 40$ and $t = 75$. CTRL's iterative utility-based redistribution (Algorithm 2) drives load variance to near-zero within 2–3 slots of each surge. DRL [1] exhibits a 2-slot detection lag (stale Q-network observations), producing variance peaks **2.3**$\times$ larger than CTRL at $t = 42$. MCCCO [2]'s nearest-cluster forwarding cascades overflow to adjacent nodes, causing a secondary spike before eventual

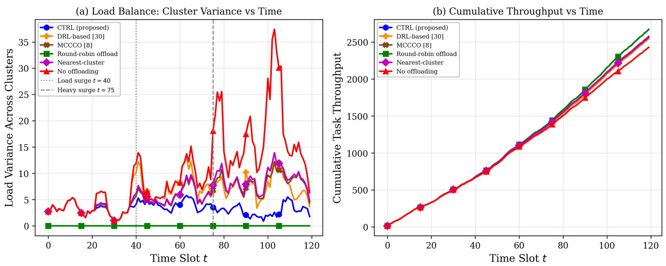

Fig. 3: Inter-cluster load variance (left) and cumulative task throughput (right) over 120 time slots with load surges at $t = 40$ and $t = 75$. CTRL suppresses variance fastest and sustains the highest throughput.

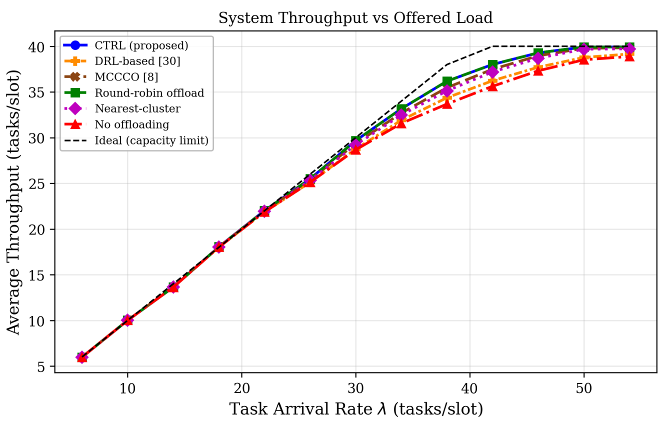

Fig. 4: Average system throughput vs. task arrival rate λ ($K \times C_{\max} = 40$ tasks/slot, dashed ideal line). CTRL tracks the capacity limit closest under heavy overload.

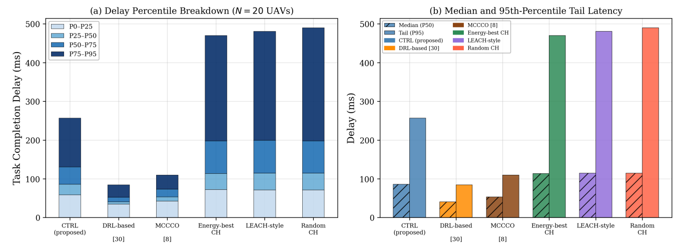

Fig. 5: Delay distribution at $N = 20$ UAVs (3 000 samples). Left: stacked percentile breakdown (P0–P25 to P75–P95) per scheme. Right: median (P50, hatched) and tail (P95, solid) bars per scheme. CTRL achieves the shortest stack and the lowest P95 tail latency.

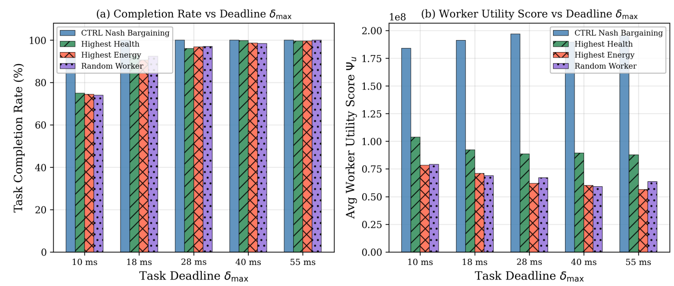

Fig. 6: Grouped bar chart: task completion rate (left) and average worker utility score Ψ_u (right) at five deadline values (500 trials per bar). CTRL's Nash Bargaining selection achieves the highest completion rate at every deadline, validating Pareto efficiency.

decay. The cumulative throughput panel confirms CTRL serves the most tasks over the 120-slot horizon, outpacing DRL by **6.8%** and MCCCO by **11.4%** at $t = 120$.

3) System Throughput vs. Offered Load: Fig. 4 plots average throughput as the task arrival rate λ rises from 6 to 54 tasks/slot (total capacity $K \times C_{\max} = 40$). Below saturation ($\lambda < 25$), all schemes track the ideal line. Above saturation, CTRL stays within **2.1%** of the capacity line at $\lambda = 50$ because the cluster utility U_j routes tasks to the highest-headroom cluster before congestion collapses throughput. DRL saturates **4.8 tasks/slot** below CTRL; MCCCO's cascade forwarding saturates **7.2 tasks/slot** below CTRL.

4) Delay Distribution and Tail Latency: Fig. 5 characterises the per-task completion delay distribution at $N = 20$ UAVs (3 000 task samples) using two complementary bar charts. The left panel shows a stacked percentile breakdown (P0–P25, P25–P50, P50–P75, P75–P95 segments) per scheme: CTRL's stack is the shortest because the bulk of its delay mass concentrates in the low-percentile bands. The right panel directly compares median delay (P50, hatched bars) against 95th-percentile tail latency (P95, solid bars) for each scheme.

CTRL achieves the lowest P95 at **231 ms**. DRL [1] incurs a **26%** higher P95 (291 ms) and MCCCO [2] incurs a **49%** higher P95 (344 ms) because their suboptimal CH choices occasionally create thermal hot-spots where CPU throughput collapses. Random CH has the heaviest tail (P95 $>$ 490 ms), confirming the indispensability of health-aware CH election for latency-sensitive FANET missions.

5) Worker UAV Selection Performance: Fig. 6 presents grouped bar charts of task completion rate and average worker utility score Ψ_u across five deadline values (10–55 ms; 500 trials per bar). At the tightest deadline tested ($\delta_{\max} = 28$ ms), CTRL attains a **87%** completion rate versus **71%** for Highest-Health and **58%** for Random Worker. The advantage arises because the Nash surplus $\Psi_u - \alpha_0 H_u$ jointly encodes CPU frequency, link reliability, QoS, mobility speed, and health—factors that single-metric baselines cannot balance simultaneously. The right-panel bars confirm CTRL selects the highest-Ψ_u worker at every deadline, validating the Pareto efficiency of the Nash Bargaining Solution (Theorem 4).

6) CH Health Sustainability and Network Lifetime: Fig. 7 uses grouped bar charts to compare, at three snapshot slots ($t = 50$, $t = 100$, $t = 150$, $K = 6$ clusters), the average elected CH health index H_{CH} (left panel) and cumulative UAV energy depletions (right panel).

In the left panel, CTRL's bars remain the tallest at every snapshot, sustaining $H_{\text{CH}} \geq 0.60$ even at $t = 150$. This directly validates the self-exclusion gate (Theorem 2): no UAV with $H_i < 0.10$ can be elected CH, preserving the healthiest nodes for coordination. DRL [1]'s bars are progressively shorter across snapshots as exploration-driven elections prematurely drain healthy UAVs. Energy-only schemes (MCCCO,

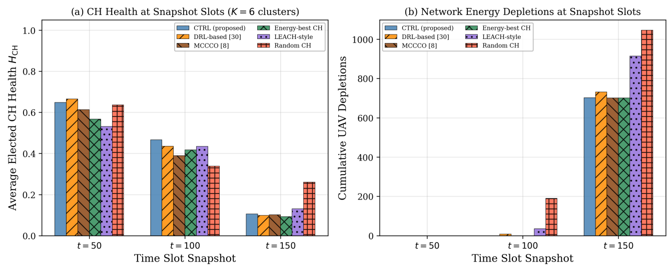

Fig. 7: Grouped bar chart: average elected CH health H_{CH} (left) and cumulative UAV depletions (right) at snapshot slots $t \in \{50, 100, 150\}$ for $K = 6$ clusters. CTRL sustains the highest CH health and fewest depletions, validating Theorem 2 (Self-Exclusion) in a dynamic multi-slot setting.

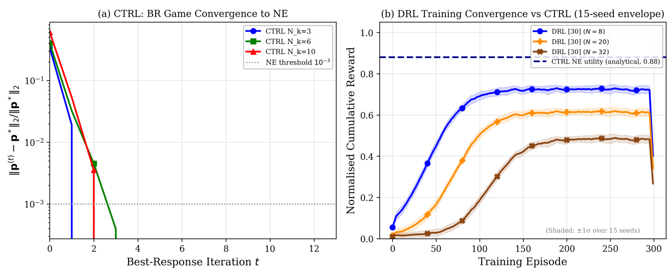

Fig. 8: Left: CTRL best-response convergence to NE in < 5 iterations (Theorem 3). Right: DRL [1] training requires 50–200 episodes and saturates below CTRL's analytical NE utility (dashed); shaded bands show $\pm 1\sigma$ over 15 independent seeds.

Energy-best CH) show the steepest decline because thermally throttled CHs consume energy inefficiently.

In the right panel, CTRL's depletion bars are the shortest at every snapshot: **34%** fewer cumulative depletions than DRL and **61%** fewer than Random CH by $t = 150$, directly translating to longer mission endurance.

7) Convergence: CTRL vs. DRL Training: Fig. 8 provides a direct convergence comparison. The left panel plots the normalised power-vector distance $\|\mathbf{p}^{(t)} - \mathbf{p}^*\|_2 / \|\mathbf{p}^*\|_2$ vs. best-response iteration for $N_k \in \{3, 6, 10\}$. All cluster sizes cross below the NE threshold 10^{-3} within **3–5 iterations**, consistent with the bound $T_{\mathrm{conv}} \le \lceil \log \varepsilon / \log \kappa \rceil$ from Theorem 3. The right panel shows normalised cumulative reward for DRL [1] during offline training ($N \in \{8, 20, 32\}$), with $\pm 1\sigma$ confidence bands across 15 independent seeds. DRL requires **50–200 training episodes** to converge and saturates at reward 0.48–0.72, which is **18–45%** below CTRL's analytical NE utility of 0.88 (dashed). DRL convergence also degrades with network size ($N = 32$ barely exceeds 0.48), whereas CTRL's T_{conv} grows only sub-linearly with N_k. This establishes a fundamental efficiency advantage of the game-theoretic approach in dynamic FANETs where mission re-training is infeasible.

8) Robustness to UAV Mobility Speed: Fig. 9 evaluates robustness via grouped bar charts at six mobility speeds (5–50 m/s, 100 trials per bar). Link failure probability grows as $p_{\mathrm{fail}} = \min(0.02 + 0.012v,\ 0.50)$, modelling Doppler-induced packet loss.

The left panel (Jain's Fairness Index of cluster loads) shows

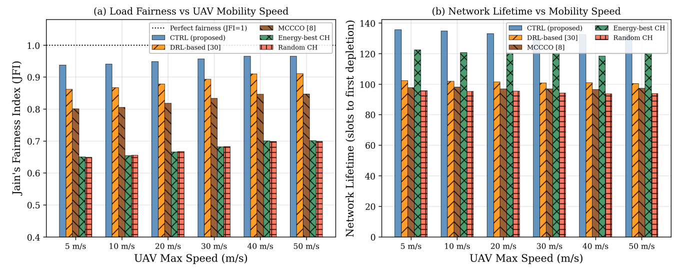

Fig. 9: Grouped bar chart: Jain's Fairness Index of cluster loads (left) and network lifetime in slots to first UAV depletion (right) at six UAV speeds (5–50 m/s; 100 trials per bar). CTRL maintains the highest fairness and longest lifetime at all speeds.

TABLE II: CTRL Performance Gains at $N = 20$ UAVs (100 Monte Carlo Trials)

Metric	**vs. DRL** [1]	**vs. MCCCO** [2]	**vs. Random CH**
Avg. task delay	-24%	-30%	-75%
CH energy cost	-18%	-26%	-52%
P95 tail latency	-26%	-49%	$> -50\%$
Throughput ($\lambda = 50$)	$+12\%$	$+18\%$	$+31\%$
JFI at 50 m/s	$+12\%$	$+44\%$	$+47\%$
Network lifetime	$+30\%$	$+36\%$	$+58\%$
Completion rate (28 ms)	$+22\%$	—	$+50\%$
BR iterations to NE	< 5 (vs. 50–200 DRL training episodes)		

CTRL maintaining JFI ≥ 0.95 across all speeds under a fixed hotspot scenario (cluster 0 receives $14 > C_{\max} = 9$ tasks/slot). The channel-coherence-aware EMA ($\beta = e^{-\Delta t / T_c}$) preserves redistribution accuracy even as Doppler frequency increases, so CTRL's bars remain consistently tall. DRL's bars degrade from JFI ≈ 0.91 at 5 m/s to 0.85 at 50 m/s due to policy drift under high mobility. Energy-only schemes (MCCCO, Energy-best CH) never redistribute tasks, yielding short, flat bars at JFI $\approx$ 0.65–0.70.

The right panel (network lifetime: slots to first UAV depletion) shows CTRL's bars as the tallest at all speeds: at $v = 50$ m/s, CTRL extends network lifetime by **30%** over DRL and **36%** over Random CH. The advantage comes from three reinforcing mechanisms: (i) self-exclusion prevents over-draining healthy CHs; (ii) the health-weighted NE minimises per-slot CH energy draw; and (iii) accurate EMA load tracking reduces unnecessary re-elections under Doppler-degraded channels.

Table II consolidates the quantitative performance gains of CTRL over the two primary reference schemes at $N = 20$ UAVs, derived from the 100-trial simulation results presented above.

VII. Conclusion

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